Problem set – 11 PAL, PLA

1. Design a BCD-to-XS-3 code converter using a (a) PROM, (b) PLA and (c) PAL
2. Design an XS-3-to-BCD code converter using a (a) PROM, (b) PLA and (c) PAL
3. Implement the following Boolean functions using PLA
4. Implement the following Boolean functions using PAL.

$$f_1(w,x,y,z) = \sum m(0,2,5,7,8,10,12,13)$$

$$f_2(w,x,y,z) = \sum m(0,2,6,8,9,14,15)$$

$$f_3(w,x,y,z) = \sum m(0,8,14,15)$$

$$f_4(w,x,y,z) = \sum m(0,1,2,4,5,8,9,10)$$

5. Tabulate the PLA programming table for the four Boolean functions listed below.

$f_1(A,B,C) = \sum m(0,1,2,4,6)$; $f_2(A,B,C) = \sum m(0,2,6,7)$

$f_3(A,B,C) = \sum m(3,6)$; $f_4(A,B,C) = \sum m(1,3,5,7)$